TEXORA · SESSIONAL NOTES · STUDY SHEET

Study of Thread Path and Main Adjustment Points of Different Industrial Sewing Machines

Threading the lockstitch and overlock machines completely, setting and balancing tensions against the bobbin reference, and locating every main adjustment point with its function and guide value.

TE-3108 · APPAREL MANUFACTURING TECHNOLOGY-II SESSIONAL · STUDY 9 · LEVEL 3 TERM 1

1. Aim and place of the study

Aim: to thread the single-needle lockstitch machine and the 3-thread overlock completely and correctly, to set needle and bobbin tensions to a balanced stitch, and to locate and state the function of the main adjustment points of both machines.

2. Materials, machine settings and specimens

Materials and machines: SNLS lockstitch machine with bobbin and case; 3-thread overlock; cones in contrasting colours (needle thread one colour, bobbin/looper threads others) so each thread's path and stitch role can be seen; test fabric (poplin), screwdriver, the machines' manuals.

Trouser measurement chart used throughout TE-3108 (men's basic trouser, garment values in cm): waist S 77 · **M 82** · L 87 · XL 92 (grade 5); hip 97 · **102** · 107 · 112 (grade 5); outseam (incl. waistband) 102 · **104** · 106 · 107; inseam 76 · **78** · 80 · 81; thigh 60 · **62** · 64 · 66 (grade 2); knee 44 · **46** · 48 · 50 (grade 2); hem (bottom) width 38 · **40** · 42 · 44 (grade 2); front rise 25 · **26** · 27 · 28. Size M is the working size unless the study states otherwise. Seam allowances, as in TE-2207/TE-2208: **1 cm** general, **1.5 cm** at side seams, waistband and other adjustment seams, **2.5 cm** at hems. The shirt values of TE-2207/2208 (size M chest 107, length 73, shoulder 46, sleeve 61.5, collar 40; grades 5/2/2/1.5/2 cm) are used unchanged wherever a shirt is the specimen.

3. The lockstitch thread path

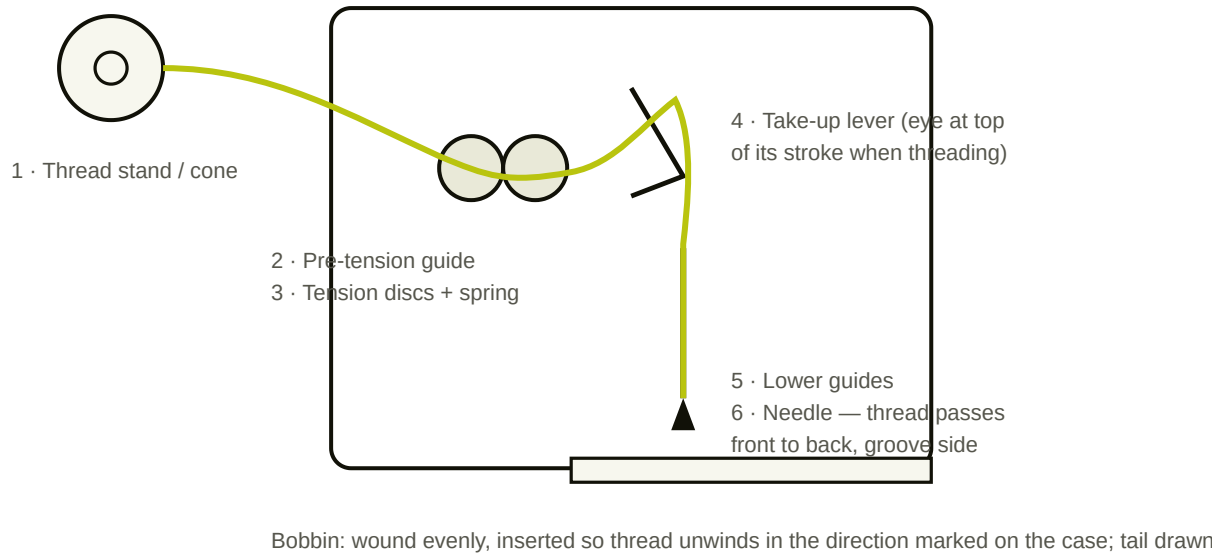


Figure 1 — Thread path of a single-needle lockstitch machine, cone to needle (green). A single missed guide changes the tension balance of every stitch.

1. Cone on the stand; thread up through the stand guide so it unwinds freely without snagging the cone base.
2. Through the machine's top guide and the pre-tension post.
3. Between the **tension discs** (thread must seat fully between them, under the take-up spring), then under/over the thread guide as the model requires.
4. Up through the **take-up lever eye** — always thread with the lever at its highest position, or the loop of thread the lever controls is lost and the first stitches break.
5. Down through the lower guides and the needle-bar guide.
6. Through the **needle eye from the groove side (front to back on most SNLS)**, leaving a tail of about 8 cm. The long groove must face the threading direction: thread running against the groove shreds.
7. Bobbin: wind evenly to just below the rim, insert into the case so it unwinds in the maker's marked direction, draw the tail through the case spring slot until it clicks under the spring; combined tails held for the first stitches.

4. Main adjustment points — lockstitch

ADJUSTMENT	EFFECT AND SETTING GUIDE
Needle-thread tension (disc nut)	Balances the stitch against the bobbin; set last, on the actual fabric, until the knot buries mid-ply.

Bobbin-case spring screw	The reference tension; a correctly set bobbin thread pulls with a firm, even drag. Set first, then leave it.
Stitch-length dial	Feed throw: set by density target — for 4 stitches/cm the length is 2.5 mm; verify by counting a sewn sample.
Presser-foot pressure	Feeding grip vs. marking/distortion; minimum that feeds without ply shift.
Feed-dog height	About 0.8–1.0 mm above the plate at its peak; low = slipping, high = marking and noise.
Needle–hook timing / loop lift	Mechanic's adjustment: hook point meets the needle scarf at 1.8–2.2 mm rise. Recognise its symptoms (Topic 4) even though students do not re-time machines.

5. The overlock thread path (504), in brief

Three threads, three paths, threaded in the maker's order — typically **lower looper, upper looper, needle** — because each later thread must lie correctly over the earlier ones at the looper eyes. Each path runs: stand guide → tension discs (looper tensions are light; the threads are laid, not pulled) → thread guides and looper eye(s) → for the needle, its own discs and eye (front to back). A correctly threaded 504 is proven only by sewing: chain off a short tail first (the stitch must form on air), then sew the test piece and check that the needle thread sits in a straight line while the loopers wrap the edge evenly (Figure of Study 2). If the chain will not form, one path is wrong — re-thread from zero rather than hunting disc by disc.

6. Balancing exercise

Sew three test seams with the contrasting threads: (1) as threaded; (2) with needle tension raised two full turns; (3) with it lowered two turns from (1). Mount all three, label the knot position of each (top surface / mid-ply / underside), and restore the balanced setting as the machine's recorded state.

Precautions

- Never thread with the machine running or the foot pedal live; switch off for needle-area work.
- Thread the take-up lever at its top position — the single commonest threading error of beginners.
- Change one adjustment at a time and sew a test after each; simultaneous changes cannot be diagnosed.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. List the lockstitch needle-thread path in order, cone to needle.
2. In the balancing exercise, seam (2) shows the knot on the top face. Explain and correct.
3. Why is the overlock threaded in a fixed order, and what is the 'chain-off' proof?
4. State the setting values: density 4/cm as a stitch length, feed-dog height, loop lift.

Answer key

1. Cone/stand guide → top guide → pre-tension → between the tension discs and take-up spring → take-up lever eye (at its highest) → lower and needle-bar guides → needle eye from the groove side.
2. Raised needle tension pulls the interlacing upward, so the bobbin thread shows as the knot on top. Correct by reducing needle tension (the bobbin stays as the reference) until the knot buries mid-ply on the test fabric.
3. Because the looper threads interlace at the looper eyes in a fixed geometry; a thread laid out of order cannot form the stitch however the tensions are set. Chaining off — a thread chain forming without fabric — proves all paths correct before material is committed.
4. Stitch length 2.5 mm; feed dog about 0.8–1.0 mm above the plate; loop lift 1.8–2.2 mm after bottom dead centre.

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